

Faithfulness and Content Selection in Long-Input Multi-Document Summarisation of U.S. Civil Rights Litigation



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Why summarization?



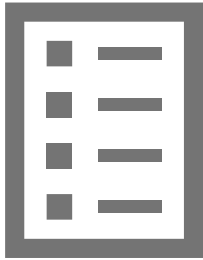
Automatic summarization of legal cases using NLP would significantly benefit legal professionals and increase **the accessibility of the law**

Summarization – Existing Methods

Extractive

vs.

Abstractive



- Most legal summarisation research
- Select key sentences and concatenate them
- Explainable
- No hallucination

- Paraphrase content
- High Performance
- Well-flowing summaries
- Not explainable/Hallucinations

Problem Description

Long Input and Multiple Documents

Average source text length of **83,340 tokens**

Mean of **10.3 documents** per case



Content Selection

Hallucinations



Evaluating Faithfulness

Research Question

Can content selection improve summary faithfulness and quality?

Dataset

Shen et al. Multi-LexSum: real-world summaries of civil rights lawsuits at multiple granularities. NIPS'22



- 4,539 **U.S. civil rights cases** (1950-2021) obtained from the Civil Rights Litigation Clearinghouse (CRLC)
- **Expert-written case summaries** in accessible language (mean 130 words)
- Writing summaries for standard cases takes **1-4 hours**, while complex cases can take **10+ hours** for an experienced lawyer

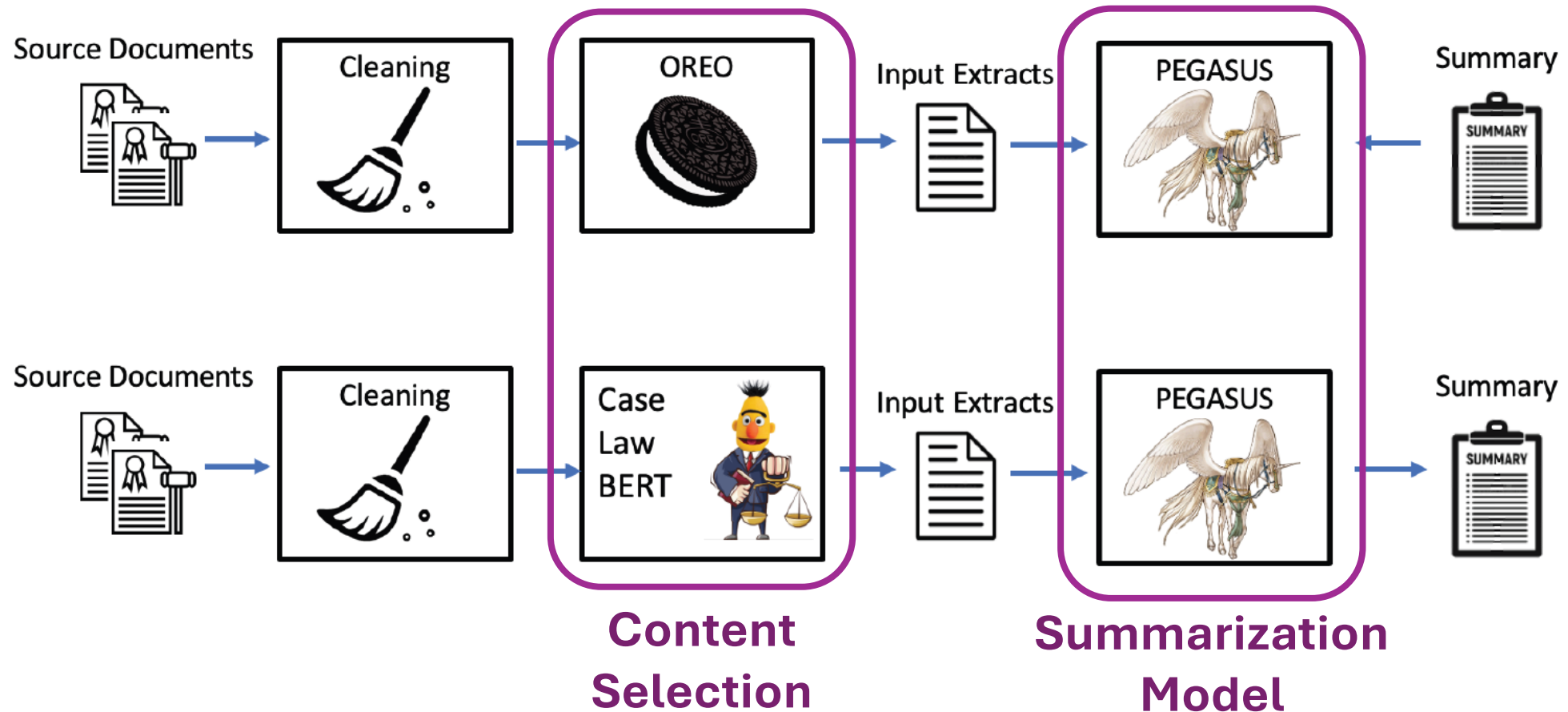
Abstractive Summarization Task

Generating a short **summary** S_i from a **set of N documents** belonging to the same case, denoted as $D_i = \{D_{i1}, D_{i2}, \dots, D_{iN}\}$.

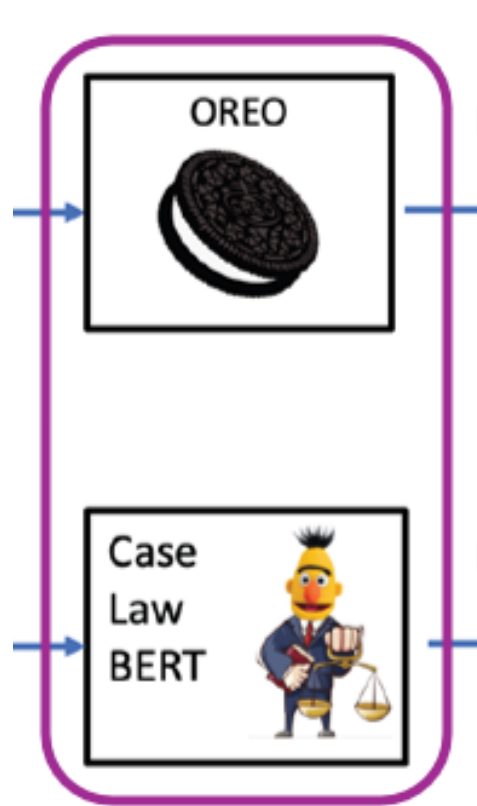
We concatenate the documents D_i for each case in **chronological order**.

Pipeline

- We vary two dimensions: **(1) Input representation**, **(2) Domain match**



Content Selection Methodology



**Content
Selection**

1

Rank input sentences

→ **Output:** A list of all source text **sentences ranked** by the classifier's confidence that the sentence contains salient information.

Testing strategies to construct the PEGASUS inputs:

- **Sentences** - we add the top-scoring sentences until the limit of 1024 tokens is reached
- **Windows** - we add the preceding and following sentences for each selected sentence
- **Paragraphs** - we add the full paragraph for each selected sentence

2

Test different context lengths

OREO: Xu and Lapata. *Text Summarization with Oracle Expectation*. ICLR 2023

Results with Content Selection

+5.56 ROUGE-1 F1

+5.46 ROUGE-2 F1

+2.70 ROUGE-L F1

+2.15 BERTScore

Comparison Legal-Pegasus OREO-Sentences to SOTA, Longformer-LED

	Method	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
Baselines	Longformer-LED (SOTA)	46.54	22.08	31.91	40.00
	PEGASUS (Reproduced)	43.23	19.26	29.35	36.15
	PEGASUS (Reported)	43.35	19.91	29.99	37.88
PEGASUS	First-1024	43.39↑	18.96↓	28.42↓	34.47↓
	First-K	43.24↑	18.96↓	28.40↓	34.94↓
	TextRank	42.36↓	17.23↓	27.31↓	33.45↓
	BERT-Sentences	43.61↑	19.33↑	27.58↓	34.52↓
	BERT-Windows	44.17↑	19.28↑	28.53↓	35.62↓
	BERT-Paragraphs	40.14↓	16.28↓	25.95↓	31.39↓
	OREO-Sentences	50.99↑	26.47↑	33.31↑	40.27↑
	OREO-Windows	47.97↑	23.28↑	31.55↑	38.92↑
Legal-PEGASUS	OREO-Paragraphs	47.15↑	22.42↑	30.83↑	37.84↑
	BERT-Sentences	42.77↓	19.08↓	27.25↓	34.81↓
	BERT-Windows	44.34↑	19.55↑	28.91↓	36.35↑
	OREO-Sentences	52.10↑	27.54↑	34.61↑	42.15↑
	OREO-Windows	48.41↑	23.72↑	31.91↑	39.44↑

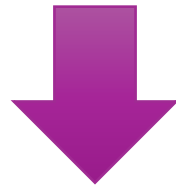
Faithfulness

Our method significantly enhances the probability that the source text entails the summary text compared to the baseline, by up to **+26.14%** (Entailment score)

Method	PEGASUS	Legal-PEGASUS
Baseline (PEGASUS reproduced)	0.2937	-
BERT-Sentences	0.5134↑	0.4954↑
BERT-Windows	0.5551↑	0.5551↑
OREO-Sentences	0.4915↑	0.4680↑
OREO-Windows	0.5457↑	0.5469↑

Key Takeaways

Long Input and Multiple Documents



Content Selection
+5.56 ROUGE-1 F1

Hallucinations



Evaluating Faithfulness
+26.14 Entailment score

- **Evidence that content selection can improve summary faithfulness and quality**
- More in the paper about failure cases and instances of hallucination